SciFig-Evaluation: A Multilingual Benchmark and Behavioural Framework for Evaluating Multimodal Language Models on Scientific Figures

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A dissertation submitted in partial fulfilment
of the requirements for the degree of
Master of Science in Artificial Intelligence
of the
University of Aberdeen.

1495

Department of Computing Science

2026

Declaration

No portion of the work contained in this document has been submitted in support of an application for a degree or qualification of this or any other university or other institution of learning. All verbatim extracts have been distinguished by quotation marks, and all sources of information have been specifically acknowledged.

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Date: 2026

Abstract

Multimodal large language models are increasingly asked to read scientific figures, yet existing benchmarks largely measure whether a model can extract a value from a chart, not whether it is a faithful visual reader. This thesis introduces SciFig-Eval, a multilingual benchmark of 1,005 figures drawn from 158 arXiv papers and paired with 1,411 expert annotations in English, Bulgarian, German and Chinese, together with a three-axis behavioural framework — Admittance, Resistance and Inductance — that separates honesty in the absence of evidence, faithfulness under misleading context, and sound inductive reasoning. Using an MQM-style multi-judge evaluation pipeline, the framework is exercised on eleven contemporary models spanning five families via five adversarial probe types. The resulting behavioural profile reveals systematic, near-orthogonal failure modes that conventional accuracy scores obscure.

Acknowledgements

I am grateful to my supervisor and to the annotation team for their time, expertise and patience. The `abdnthesis.cls` class was developed at the University of Aberdeen.

Contents

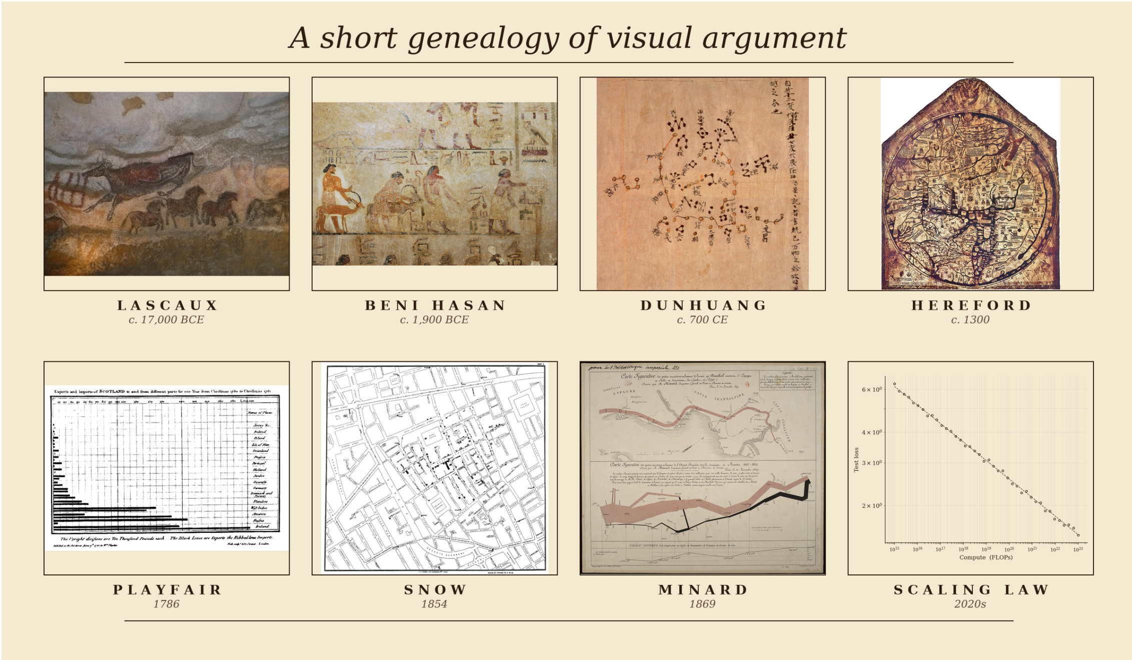
Chapter 1

Introduction

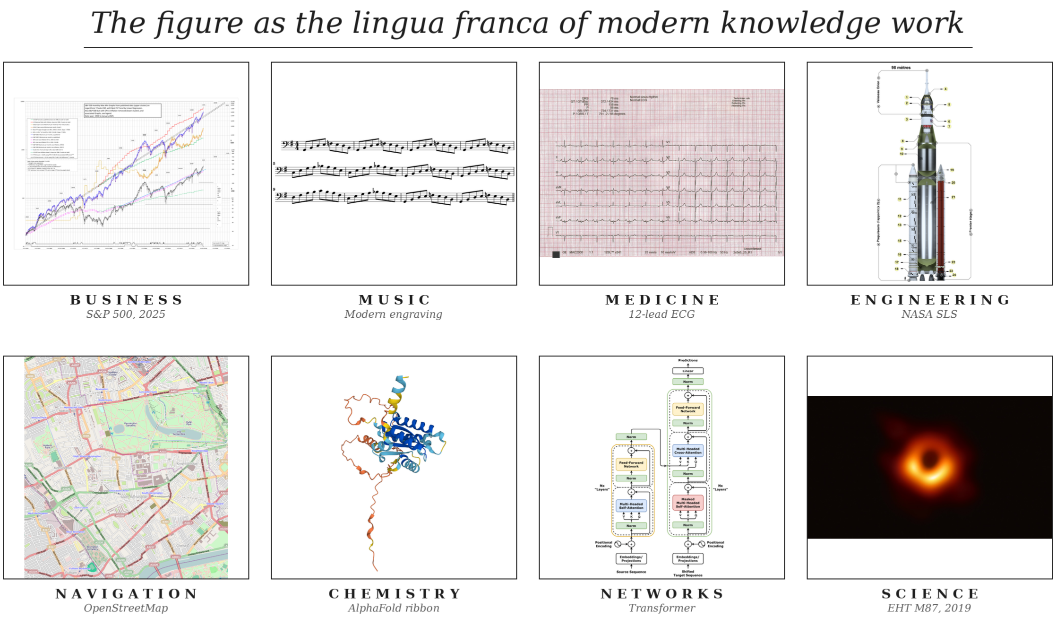
1.1 *Homo sapiens*: A Species of Visual Thinkers

The record of a species willing to reason pictorially begins no later than the Upper Palaeolithic. Some time between seventeen and fifteen millennia before the present, an inhabitant of what is now the Dordogne crushed haematite between stones, lit a pine torch, and rendered an aurochs on Lascaux limestone at approximately life size (Aujoulat, 2005). These visual arguments have continued to evolve. Twelfth-dynasty Egyptian painters at Beni Hasan committed a caravan of Semitic travellers, traditionally associated with the Joseph narrative, to a tomb wall around 1900 BCE (Kamrin, 2009); seventh-century Chinese astronomers committed over fourteen hundred stars to silk at Dunhuang (Bonnet-Bidaud et al., 2009); medieval cartographers compressed theology, geography and chronology onto the Hereford *mappa mundi* (Harvey, 1996). In 1786 William Playfair invented the bar chart in order to argue, graphically and in print, about Britain's trade balances (Playfair, 1786). John Snow's 1854 cholera map of Soho laid down the method of modern epidemiology (Johnson, 2006), and Charles Joseph Minard's 1869 flow map of Napoleon's 1812 Russian campaign (Friendly, 2002) is what Tufte (2001) has called, without serious contest, the densest argument ever made on paper (Figure 1.1a). When a human being intends a claim of any real quantitative weight, the first move is almost invariably to draw.

In contemporary practice, figures have not only evolved but have become significantly embedded in daily operations. Businesses read through dashboards of candlestick charts and cohort curves (Nison, 2001) that senior management consults in direct preference to prose; engineers coordinate the hundreds of thousands of components that constitute a ship, a hospital or an aircraft through hierarchies of blueprints read without ambiguity by people who may never speak to the designer (Ferguson, 1992); and even life-and-death decisions are made by reading a heartbeat off the grid of an electrocardiogram, or tissue contrast off a radiograph or perfusion map (Goldberger et al., 2017). Music (Treitler, 1984), satellite navigation (Kaplan and Hegarty, 2017), chemistry (Hoffmann, 1995) and network science (Newman, 2010) sustain the same diagrammatic primacy, and every competent empirical paper in the natural and social sciences eventually commits its argument to a figure (Figure 1.1b). **The figure has become the lingua franca of modern knowledge work**, and in many domains the argumentative weight it now carries is strictly greater than the weight carried by the accompanying prose.



(a)



(b)

Figure 1.1: **(a)** A short genealogy of visual argument, from Lascaux and Egyptian wall painting through the medieval *mappa mundi*, Playfair's 1786 bar chart and Snow's 1854 cholera map to modern neural scaling curves. **(b)** The figure as lingua franca of modern knowledge work across business, music, medicine, engineering, navigation, chemistry, networks and the natural sciences. *Sources.* **(a)** Lascaux aurochs, Beni Hasan caravan, Dunhuang star chart, Hereford *mappa mundi*, Playfair 1786 bar chart, Snow 1854 cholera map and Minard 1869 flow map of Napoleon's Russian campaign, all public domain (via Wikimedia Commons, the Metropolitan Museum of Art, the British Library, Hereford Cathedral, the Internet Archive and the Bibliothèque nationale de France respectively); scaling curves adapted from Kaplan et al. 2020. **(b)** A candlestick chart, score excerpt, ECG trace, blueprint, nautical chart, molecular diagram, network graph and scientific plot, from public-domain or openly-licensed repositories (Wikimedia Commons, NOAA, PhysioNet) or rendered by the author.

1.2 Can *Intelligentia Machinae* (Machine Intelligence) Comprehend Visuals Similarly?

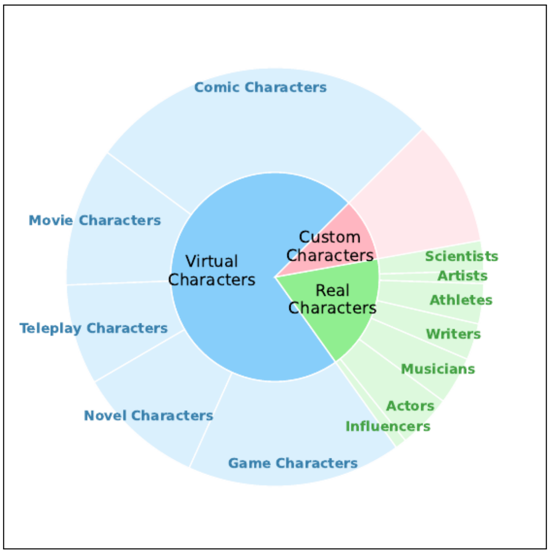
Recent progress in artificial intelligence has been uncommonly rapid, and the current generation of large models has brought within plausible reach an older ambition of the field: a machine that can collaborate with a human being to achieve a shared task. A necessary condition of that collaboration is that the two parties reason on similar planes and navigate near-similar cognitive paradigms. Exact cognitive parity is not required; only a proximity close enough to sustain useful work in concert with a human collaborator. Since the human, as the previous section establishes, is a visual thinker, that proximity obliges the machine to comprehend the visual: to read, interpret and reason from the figure with fidelity comparable to its reading of text.

Systems at the current frontier, including Anthropic's Claude Opus 4.6 and 4.7 (Anthropic, 2026a,b), OpenAI's GPT-5 series (OpenAI, 2025a) and Google DeepMind's Gemini 3 Pro (Google DeepMind, 2026a), are now deployed as infrastructure rather than laboratory curiosities. They are shipped into enterprise software through the major cloud providers, and they are measurably altering how code is written and how commercial operations are conducted. The salient question is therefore no longer whether such systems are capable, but whether that capability extends, in the way the human partner's does, to the visual.

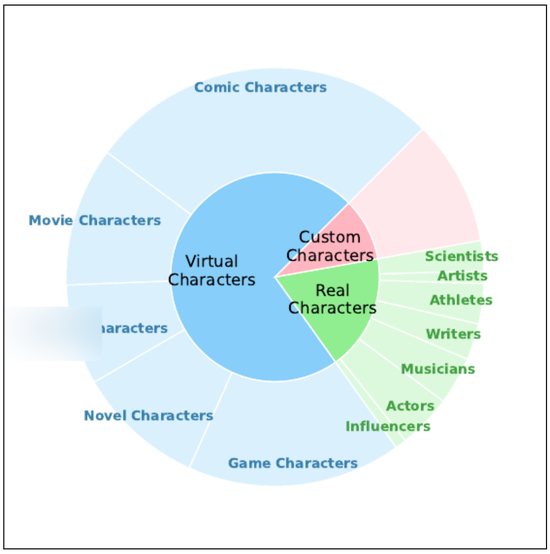
1.3 Motivation for Focusing on Scientific Research Figures

Research is among the first domains in which collaborative intelligence is being tested at scale: between late 2024 and early 2026 the three largest model developers each shipped autonomous research agents (Google, 2024; OpenAI, 2025c; Anthropic, 2025), and scientific deployments such as the Biomni biomedical platform now report research-cycle compressions of an order of magnitude or more (Anthropic, 2026d). Yet scientific knowledge is not monolingual; a substantial share of the record is produced in Chinese, German, Bulgarian and other languages of research, and the cost of excluding that share, in citation, uptake and participation, has been documented at length (Amano et al., 2023). Chart- and figure-understanding benchmarks, however, remain predominantly English: recent multilingual efforts such as PolyChartQA (Li et al., 2025), PM4Bench (Gao et al., 2025) and IndicVisionBench (Sharma et al., 2025) each report substantial cross-lingual degradation on tasks the same models solve comfortably in English. The reach of a multimodal system, considered as a research collaborator, therefore scales with the breadth of languages over which its visual reading remains reliable, and the scientific figure, considered across languages, is the natural object at which to test whether the current frontier reads visually or merely appears to. What that reading must withstand, beyond correctness on the clean image, is illustrated in Figure 1.2.

The failure is not a miscount or a mislabelled axis; it is a confident substitution of a plausible category for evidence the model cannot actually see. Honest reading, not just accurate reading, is therefore the object of interest, and the research questions below are framed accordingly: they ask how faithfully today's frontier models describe scientific figures across languages; how their reading survives visual degradation and adversarial context; and whether, when the evidence is not there, they acknowledge it rather than fill it in.



(a) Original figure.



(b) One outer label selectively blurred.

Figure 1.2: A sunburst taken from an arXiv paper in the SCIFIG-EVAL corpus. In **(a)**, the outer-ring category between *Movie Characters* and *Novel Characters* is legibly *Teleplay Characters*; in **(b)**, that single label has been blurred. Asked which category sits between the two, GPT-5.2 answers "*Anime Characters*," with no admission that the relevant region is unreadable. Source image reproduced from arXiv:2505.23923v1 under its open-access licence; blurred variant generated by the SCIFIG-EVAL pipeline.

1.4 Research Questions

The central objective of this thesis is to characterise, across models, languages and adversarial conditions, how faithfully today's frontier multimodal models actually read scientific figures, and to build an evaluation framework that surfaces the failure modes which conventional accuracy benchmarks obscure. Concretely, the investigation is organised around the following five research questions, each pairing a capacity with the conditions under which that capacity is tested:

RQ1 (Description accuracy across languages). To what extent do current vision–language models produce accurate and complete descriptions of scientific figures across typologically diverse languages?

RQ2 (Robustness to visual degradation). How does visual degradation, including noise, compression, aspect distortion and contextual embedding within paper pages, affect the accuracy (whether what is stated is correct), completeness (whether what is visible is reported) and clarity (whether what is reported is expressed unambiguously) of VLM-generated figure descriptions?

RQ3 (Comprehension beyond description). Beyond surface-level description, can VLMs demonstrate genuine comprehension of scientific visualisations, from answering quantitative questions to performing non-trivial inductive reasoning that requires distinguishing analytical insight from surface-level pattern matching?

RQ4 (Adversarial robustness). How robust are VLM outputs when subjected to adversarial manipulations, including misleading captions, contradictory premises and psychologically-informed sycophancy probes?

RQ5 (Calibrated epistemic honesty). When confronted with incomplete or unresolvable visual information, do VLMs exhibit calibrated epistemic honesty, acknowledging uncertainty rather than confabulating plausible but ungrounded responses?

1.5 Thesis Structure

The remainder of the thesis is organised as follows. Chapter 2 (Background) provides the technical and conceptual foundation: how frontier multimodal models convert figures into internal representations, how scientific figures carry argument rather than merely depict it, and the failure modes and evaluation traditions on which the remainder draws. Chapter 3 (Literature Survey) situates the work against chart understanding, multimodal hallucination, multilingual evaluation and MQM-style scoring. Chapter 4 (Design) sets out the SciFig corpus, the A–R–L framework and the five adversarial probe families. Chapter 5 (Implementation) describes the evaluation pipeline, the model and judge harness and the reproducibility controls. Chapter 6 (Evaluation) reports the main quantitative results across the 11 evaluated models and the four languages, together with ablations on judge ensembles and probe families. Chapter 7 (Discussion) interprets the findings, connects them to the A–R–L framework and acknowledges the limitations of the study. Chapter 8 (Conclusion) summarises the contributions, revisits the research questions and outlines directions for future work.

Chapter 2

Background

This chapter lays the technical ground for the behavioural evaluation developed in Chapters 4–6, organised around four questions. Section 2.1 asks how today's Large Language Models see. Section 2.2 asks how they reason about what they see, and whether that reasoning survives in non-English figures. Section 2.3 asks how they behave when the visual evidence is incomplete, misleading or contested. Section 2.4 asks how the resulting proficiency is measured at scale. The scientific figure as a carrier of argument is taken for granted from the motivation in Chapter 1, and the visualisation-grammar literature is surveyed in Chapter 3.

2.1 How do today's Large Language Models see?

Every family evaluated in this thesis shares the skeleton sketched in Figure 2.1. A raster image is partitioned into patches by a vision encoder, the resulting embeddings are retuned by a connector into the token space of a language backbone, and the backbone consumes those projected vision tokens as a prefix alongside the text prompt (Liu et al., 2023; Alayrac et al., 2022; Li et al., 2023a). From the backbone's point of view, the image arrives as a sequence of soft tokens drawn from the same vector space as text.

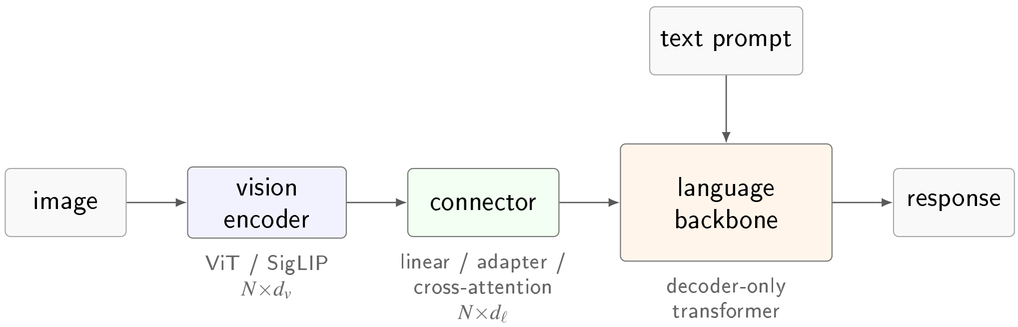


Figure 2.1: The canonical vision-language skeleton shared by every family evaluated in this thesis. A Vision Transformer (Dosovitskiy et al., 2021), typically trained with a CLIP (Radford et al., 2021) or SigLIP (Zhai et al., 2023) contrastive objective, maps the image to N patch embeddings of width d_v; a connector (linear projection, adapter, or cross-attention) retunes them to the backbone width d_ℓ (Liu et al., 2023; Li et al., 2023a); the decoder-only language model consumes them as a prefix alongside the text prompt.

Within this skeleton, the seven families evaluated here diverge along four axes that turn out to matter empirically. The vision encoder is almost always a Vision Transformer (Dosovitskiy et al., 2021) trained with a CLIP (Radford et al., 2021) or SigLIP (Zhai et al., 2023) contrastive objective, except for Qwen3-VL (Qwen Team, Alibaba, 2025; Bai et al., 2024), which trains its own from

Table 2.1: Publicly documented vision-language architecture across the seven families evaluated in this thesis. "–" denotes a detail that is not disclosed in the primary source. The four open-weight families (top block) publish detailed technical reports; the three closed frontier families (bottom block) publish system cards that address capability and safety but leave the vision stack undisclosed. *Sources.* Gemma 3 (Gemma Team, Google DeepMind, 2025); Qwen3-VL (Qwen Team, Alibaba, 2025), with architectural continuity from Qwen2-VL (Bai et al., 2024); Llama 4 (Meta AI, 2025); Phi-4 multimodal (Microsoft Research, 2025); Claude Opus 4.x (Anthropic, 2026c); GPT-5 / 5.2 (OpenAI, 2025b); Gemini 3 Pro (Google DeepMind, 2026b).

Family	Vision encoder	Resolution / tiling	Image tokens	Fusion regime
Gemma 3	frozen SigLIP-400M	896×896 + Pan&Scan	256 (linear proj.)	adapter
Qwen3-VL	custom ViT + 2D-RoPE	dynamic, any aspect	variable (MRoPE)	native
Llama 4	MetaCLIP-derived ViT	–	– (early-fusion)	native early-fusion
Phi-4 mm	CLIP variant + LoRA	–	–	Mixture-of-LoRAs
Claude Opus 4.x	–	$\leq$ 2576 px long edge	–	–
GPT-5 / 5.2	–	–	–	native (claimed)
Gemini 3 Pro	–	up to 900 images / prompt	–	native (sparse MoE)

scratch with 2D-RoPE to accept arbitrary resolution. The connector is either a linear projection (Liu et al., 2023), a cross-attention adapter with learnable queries (Alayrac et al., 2022; Li et al., 2023a), or LoRA modules grafted into the backbone itself (Hu et al., 2022; Microsoft Research, 2025). The fusion regime ranges from late adapter onto a frozen backbone (Gemma 3, LLaVA), through early-fusion on interleaved tokens (Meta AI, 2025; Chameleon Team, 2024), to ground-up native co-training (Qwen3-VL and the closed frontier families). The per-image token budget, finally, varies from a fixed 256 (Gemma Team, Google DeepMind, 2025) to aspect-preserving variable-length sequences (Bai et al., 2024). Table 2.1 collects what each family publicly discloses along these axes.

The table is as informative for what it omits as for what it contains. The four open-weight families each publish a technical report that documents the vision stack end to end; the three closed frontier families publish system cards (Anthropic, 2026c; OpenAI, 2025b; Google DeepMind, 2026b) oriented toward capability and safety, and leave the encoder, the resolution policy, the token budget and the connector architecture undisclosed. The asymmetry is structural. The models most widely deployed in real research settings are the models whose visual reading is architecturally least inspectable, and an evaluation that required internal access would, at the frontier, have almost nothing to evaluate. This thesis therefore treats that constraint as a design requirement, and Chapter 4 characterises each model's visual reading from the outside, through controlled perturbations of the input and the text-in, text-out interface that every family exposes.

2.2 How do today's LLMs reason about what they see, and is it impacted by language?

Vision-language benchmarks distinguish description from reasoning. Description asks what is on the page (the value at a tick, the colour of a series, the label of a cluster); reasoning asks what follows from it (whether a trend is monotonic, whether two series diverge, whether a stated conclusion is supported by the plotted data). CharXiv (Wang et al., 2024) splits its questions along exactly this line, and the gap between the halves is substantial on every frontier model. ChartMuseum (Tang et al., 2025) sharpens the pattern on harder real-world charts, where the best open model scores 38.5%, the best closed model 63%, and humans above 90%.

The conventional reading blames the reasoner, but recent diagnostics locate the bottleneck earlier. Before the backbone can infer anything the encoder has to have resolved the tick, the legend key or the axis label, and in practice it often has not. Controlled ablations attribute the majority of the residual error to perception rather than inference, and replacing the raw chart with its ground-truth table lifts performance sharply even when the nominal question is a reasoning one (Wang et al., 2024; Tang et al., 2025). The visual reading, not the deductive capacity, is the first-order limit.

Language multiplies this cost. Chinese, German, Bulgarian and other languages host substantial shares of the primary literature, and the price of excluding them in citation, uptake and collaboration has been documented at length (Amano et al., 2023). Chart-understanding benchmarks remain predominantly English, and recent multilingual efforts (PolyChartQA (Li et al., 2025), PM4Bench (Gao et al., 2025), IndicVisionBench (Sharma et al., 2025)) each report substantial cross-lingual degradation on tasks the same models solve comfortably in English. PM4Bench further shows that higher-level reasoning scores scale with lower-level OCR accuracy, consistent with the perception-bottleneck account.

Reading a non-English figure stresses the model at three layers. At the *pixel* layer the encoder must resolve non-Latin script (Cyrillic superscripts, Chinese logographic axis labels, German compounds that sit poorly in a tight tick space) through the same patch grid it was trained on for English. At the *token* layer the backbone's tokeniser fragments non-English strings into finer subword units, attenuating any learned lexical prior. At the *prior* layer the backbone's world knowledge is itself skewed toward English-dominant pre-training, so that the completion a confident model gravitates toward is systematically more likely in English than in Bulgarian. SCIFIG-EVAL probes each layer through Bulgarian (Slavic, Cyrillic), German (Germanic, Latin script with heavy compounding) and Chinese (Sinitic, logographic), spanning script families, morphological complexity and pre-training density.

2.3 How do today's LLMs behave under difficult or adversarial seeing conditions?

Three patterns of failure recur once the image is degraded, occluded or framed by misleading text, and each is observable only through controlled perturbation of the input.

The first is confabulation under missing evidence. Asked about a region that is blurred, cropped or simply absent, a vision-language model will often produce a confident, plausible-sounding answer rather than acknowledge the gap. POPE (Li et al., 2023b) quantifies this at the object level, HallusionBench (Guan et al., 2024) extends it to figure-level visual-reasoning traps, and ChartHal (Xu et al., 2025) documents the same pattern on charts through controlled visual ablations in which the model is asked about content the benchmark has removed. Figure 1.2 in Chapter 1 illustrates the phenomenon, where GPT-5.2 answers "Anime Characters" about a blurred sunburst label as though it were still legible.

The second is drift under misleading context. Sycophancy, the tendency to agree with an interlocutor's implied position even when it is false, was formalised by Sharma et al. (2023) in language models, and its visual analogue is now documented by Zhao et al. (2024). VLMs shift their description of a figure in response to leading questions, misleading captions or asserted prior beliefs, sometimes contradicting their own previous output on the same image. The closely related textual-dominance effect, in which the model follows an assertion stated in the prompt even when the image contradicts it, carries the same signature. The textual channel routinely overrides what the encoder has actually reported.

The third is confident inference past the available evidence. Kadavath et al. (2022) showed that language models can, under the right conditions, know roughly what they know; the visual setting has proven harder. Recent studies tie object hallucination directly to high epistemic uncertainty on early vision-encoder tokens and to disproportionate attention on a small number of "blind" patches, so that what looks from the output like careless reasoning is often a quiet perceptual failure at the first layer. Calibration, on this view, is not a separate accuracy metric but the substrate on which honest abstention rests.

These three patterns describe a technical surface that an accuracy benchmark cannot probe, because each can coexist with a high score on the underlying clean-image task. They motivate, and are operationalised as, the behavioural framework that Chapter 4 develops.

2.4 How do we evaluate the seeing proficiency?

Evaluating 11 models across four languages, five probe families and 1,005 figures requires a rubric richer than accuracy on a single-answer question, and an apparatus that can apply it without human graders in the loop.

Chart-understanding benchmarks have moved in this direction over the past decade. The first generation (PlotQA (Methani et al., 2020), ChartQA (Masry et al., 2022)) paired largely synthetic charts with short-answer questions amenable to exact-match grading. The second (CharXiv (Wang et al., 2024), ChartQAPro (Masry et al., 2025)) moved to real scientific charts and split questions into descriptive and reasoning subsets. The most recent (ChartMuseum (Tang et al., 2025), ChartHal (Xu et al., 2025)) drop exact-match in favour of open-ended description, reasoning and controlled abstention, the form this thesis evaluates.

Open-ended outputs are not scorable by exact match. Multidimensional Quality Metrics (MQM), introduced by Lommel et al. (2014) for machine translation, supply the rubric this thesis adapts. MQM replaces a scalar with an error typology (accuracy, fluency, terminology, style) in which each error carries a severity weight (minor, major, critical), and aggregates the resulting annotations into a calibrated quality score. The same apparatus transfers to figure description, where the categories become visual fidelity, completeness and clarity, and severity separates a misread tick from a fabricated series.

At benchmark scale, human MQM annotation is infeasible, and the literature has converged on the *LLM-as-judge* paradigm (Zheng et al., 2023). The approach is fast, reproducible and, when carefully deployed, reaches human-level agreement. It is also subject to well-documented biases, including self-preference (Wataoka et al., 2024), position bias in pairwise comparisons (Zheng et al., 2023), and length, verbosity and concreteness biases in scoring. The established mitigations (randomising positions, averaging over swaps, and aggregating across a heterogeneous judge ensemble) are adopted wholesale in Chapter 5. The pipeline used here draws from three judge families (Claude Opus 4.x, GPT-5 / 5.2 and Gemini 3 Pro) to dilute any single family's preferences, and scores every candidate twice per rubric cell to cancel position bias. Together with the behavioural probes of §2.3, this apparatus furnishes the evaluation harness for Chapters 5 and 6.

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